FitBuddy-AI Fitness Plan Generator using Gemini Models

Phase 7 – Project Documentation

1. Introduction

The documentation phase records the project in a structured form so that another person can understand its purpose, architecture, implementation, testing, and operation. Good documentation is particularly important for FitBuddy because the project contains several technologies and multiple AI-related modules.

2. Project Overview

FitBuddy is an AI-powered fitness plan generator. The application collects basic user information and uses Gemini-based generation to produce a personalized workout plan. It also generates a nutrition tip and stores the information in a database. Users can provide feedback and request an updated plan.

The application is implemented as a web-based Python project using FastAPI. HTML pages are rendered with Jinja2 templates. SQLite and SQLAlchemy are used for storage, and the Google GenAI/Gemini SDK is used for AI integration.

3. Technology Stack

Python is the core development language. FastAPI provides API routing and request handling. Uvicorn is the ASGI server used to run the application. SQLAlchemy provides database abstraction and SQLite provides local storage. Jinja2 is responsible for server-side HTML templates. HTML and CSS provide the user interface. python-multipart supports form submissions, while

dotenv/configuration support allows environment-based settings. Gemini provides the generative AI functionality.

4. Project Modules

The documentation should describe each major file and its purpose. main.py initializes the application. routes.py contains the endpoints. database.py manages storage and the User model. schemas.py contains structured request models. config.py contains configuration values.

The AI package contains the workout generator, nutrition generator, and updated-plan module. The templates directory contains the main input, result, and administrator pages. This organization makes it easier to locate functionality and explain the project during review.

5. Complete User Workflow

The documented workflow begins when the user opens the home page. The user enters username, user ID, age, weight, goal, and intensity. After submission, the backend validates the information, generates a workout plan, generates a nutrition tip, and saves the result.

The result page presents the generated information. If the user submits feedback, the backend retrieves the stored user, sends the original plan and feedback to the update module, saves the revised plan, and displays the updated result. The administrator workflow allows authorized viewing and deletion of users.

6. API Documentation

FastAPI automatically provides interactive API documentation through the /docs endpoint. The current application exposes routes for the home page, workout generation, feedback submission, viewing all users, deleting a user, and checking service health.

The Swagger interface is useful during development because it displays the available endpoints and the expected parameters or request bodies. It can also be used to demonstrate that the backend is correctly registered.

7. Installation and Execution Notes

The project should be run from the project directory with the virtual environment activated. Required packages must be installed in that environment. The application is then started with Uvicorn in reload mode during development. The browser can access the local server address, while /docs can be used for API documentation.

The documentation should also record common setup issues, including missing dependencies, incorrect module imports, missing template/static directories, and incorrect template response configuration. Recording these issues helps future developers troubleshoot the project.

8. Conclusion

The documentation phase creates a complete reference for FitBuddy. It records the project purpose, technologies, modules, workflow, API endpoints, execution method, and important development notes. This document can be used during internal review, project submission, viva, and demonstration.

Phase Deliverables / Evidence

Item	Purpose	Status / Evidence
Project overview	Explains system purpose	Documented
Technical reference	Explains modules and workflow	Documented
Execution notes	Explains running and troubleshooting	Documented